

TRANSPLANTATION

Mucosal calprotectin is associated with severity of aGI-GVHD and poor outcomes after allogeneic stem cell transplantation

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KEY POINTS

- Mucosal calprotectin and neutrophil infiltration correlate with aGI-GVHD severity.
- Small and large intestines display distinct and site-specific pathophysiology.

Calprotectin, a calcium- and zinc-binding protein that comprises the subunits S100A8 and S100A9, has been extensively studied as a biomarker of gastrointestinal (GI) inflammation through fecal and serum analyses. However, its role in intestinal tissue remains poorly understood because of the limited availability of biopsy specimens. In this study, we analyzed S100A8 and S100A9 messenger RNA (mRNA) expression in 579 intestinal biopsy specimens from allogeneic stem cell transplantation recipients and observed a strong association with acute GI graft-versus-host disease (aGI-GVHD; $P < .001$). Neutrophil infiltration correlated with the severity of aGI-GVHD ($P < .001$), and calprotectin expression was strongly linked to Toll-like receptor 4 (TLR4; $P < .001$) and TLR2 ($P < .001$) expression. Both TLR4 and aGI-GVHD were associated with elevated calprotectin mRNA levels ($P < .001$). When patients received broad-spectrum antibiotics at disease onset, calprotectin expression was suppressed (S100A8, $P = .001$; S100A9, $P = .01$).

GI site-specific differences in calprotectin expression were identified: during severe aGI-GVHD, levels increased up to 30-fold in the small intestine and up to fivefold in the large intestine with respect to mild or no aGI-GVHD, whereas under homeostasis, the large intestine exhibited higher baseline calprotectin ($P = .001$). The high clinical relevance of this finding is evident from the observation that calprotectin expression was prognostic for transplant-related mortality. Our study suggests that (1) calprotectin is a potential biopsy biomarker in aGI-GVHD and (2) calprotectin expression and neutrophil infiltration possibly indicate translocation of microbiota, which (3) may be modulated by antibiotics.

Introduction

Allogeneic stem cell transplantation (ASCT) is a potentially curative treatment for a range of malignant and nonmalignant conditions.¹ However, its efficacy is significantly constrained by acute graft-versus-host disease (aGVHD) and chronic GVHD, arising from donor cells targeting the host's healthy tissues.²⁻⁴ aGVHD, the most common and life-threatening complication, affects nearly 50% of transplant recipients, with the involvement of skin (81%), gastrointestinal (GI) tract (54%), and liver (50%) as affected organs.³

Despite advancements in HLA typing, conditioning regimens, and GVHD prophylaxis,⁵ acute GI GVHD (aGI-GVHD) remains a leading cause of transplant-related mortality (TRM). Clinically, aGI-GVHD is characterized by severe abdominal pain, (bloody) diarrhea, nausea, and anorexia; its histopathological hallmarks include tissue damage, crypt loss, and epithelial denudation.⁶ Upon tissue injury, innate immune cells from the myeloid lineage, particularly neutrophils, macrophages, and monocytes, are the primary responders. Among these, neutrophils, the most abundant leukocytes in circulation, are the first to arrive at injured tissues.⁷ Neutrophil recruitment to sites of

infection or injury is mediated by a multistep process involving selectins, integrins, and chemokines.⁸ Notably, neutrophil-derived calprotectin, a potent antimicrobial and antifungal protein,⁹ has been increasingly implicated in GI inflammatory conditions such as inflammatory bowel disease,^{10,11} sepsis,¹² necrotizing enterocolitis,¹³ aGI-GVHD,¹⁴⁻¹⁷ as well as pulmonary infections like coronavirus disease 2019.^{18,19}

Calprotectin, a calcium-binding S100 leukocyte protein, is evolutionarily conserved and comprises 2 subunits in mammals: S100A8 and S100A9, also known as myeloid-related proteins 8 and 14, respectively.²⁰ Originally identified in the 1980s as leukocyte protein L1, calprotectin was abundantly observed in granulocytes.²¹ Its calcium-binding properties led to its renaming as "calprotectin."²² Representing nearly half of the cytosolic protein in neutrophils, calprotectin demonstrates a 40-fold higher expression in neutrophils than in monocytes.²³ Calprotectin functions as an endogenous activator of Toll-like receptor 4 (TLR4), amplifying inflammatory responses via the TLR4–myeloid differentiation factor 2 (MD2) pathway.²⁴ In the specific context of intestinal GVHD, fecal and/or serum calprotectin has been established as both a diagnostic biomarker for assessing disease severity and a prognostic biomarker for predicting clinical outcomes.^{14,25-27}

The GI mucosa, when damaged, serves as a major site of neutrophil-derived calprotectin. In the context of aGI-GVHD, bacterial translocation (BT) from the lumen to the mucosa through the compromised epithelial barrier is a likely phenomenon. This translocation may trigger neutrophil infiltration into the damaged tissue,^{28,29} during which neutrophils release calprotectin in response to pathogen-associated molecular patterns and danger-associated molecular patterns.^{30,31} Released calprotectin subsequently (1) enters systemic circulation and (2) is secreted into the gut lumen due to epithelial damage. These dynamics support the diagnostic potential of serum and fecal calprotectin as noninvasive biomarkers, which have been extensively studied. However, mucosal tissue, the primary site of inflammation, is less frequently analyzed because of the invasive nature of the sampling process.

In this retrospective study, we investigated the extent of neutrophil infiltration and calprotectin expression in mucosal biopsy specimens obtained from ASCT recipients with and without aGI-GVHD. Our findings revealed a strong correlation between neutrophil infiltrates and the subunits of calprotectin. Notably, mucosal calprotectin expression was elevated in the presence of aGI-GVHD and showed a positive association with disease severity. Furthermore, both aGI-GVHD and TLR4 were associated with elevated calprotectin levels. Interestingly, significant differences in calprotectin expression were observed between the small and large intestines, underscoring potential site-specific pathophysiological mechanisms. Intriguingly, when patients received broad-spectrum antibiotics at the onset of aGI-GVHD, the expression of calprotectin was suppressed effectively.

Materials and methods

Patient characteristics

All patients gave informed consent, and the study was approved by the local ethical review board Ethikkommission

der Universität (approval numbers 02/220, 09/059 and 17-619-101). All the studies were performed according to the Declaration of Helsinki. A total of 309 patients who underwent ASCT in the University Hospital Regensburg, Regensburg, Germany, from July 2005 to July 2022 were included in the study. Detailed patient characteristics can be found in supplemental Table 1, available on the *Blood* website.

Biopsy characteristics

A total of 579 retrospective biopsy specimens were included in this study. The median interval from ASCT to biopsy specimen collection was 50 days (range, 13-4795), with 75.9% of specimens collected within 6 months after ASCT. Mucosal biopsy specimens were taken under 3 conditions: (1) as part of a screening study in asymptomatic patients ($n = 225$; median, 30 days), (2) in response to clinical symptoms suggestive of de novo onset ($n = 181$; median, 27 days), and (c3) for ongoing GVHD cases involving persistent or recurrent disease ($n = 173$; median, 192 days). Anatomical sites for biopsy specimen included the upper GI tract (UGI; $n = 218$; median, 42.5 days) and lower GI tract (LGI; $n = 361$; median, 53.5 days). To minimize bias from multiple biopsy specimens in patients with severe disease courses, we identified a single "master" biopsy per patient, defined as the most clinically relevant specimen for outcome analyses.

RNA extraction, cDNA synthesis, and qPCR

Upon retrieval, biopsy specimens were immediately transferred to 500 μ L RNA protect (Qiagen) and were stored at -80°C until RNA extraction. RNA extraction, including DNA digestion, was performed with RNeasy Mini Kit as per manufacturer's instructions. RNA concentration and purity were determined by Nanodrop and Bioanalyzer, respectively. One microgram of RNA was reverse transcribed to complementary DNA (cDNA) using Moloney murine leukemia virus reverse transcriptase (Promega) according to the manufacturer's instructions. Quantitative polymerase chain reaction (qPCR) was performed on a LightCycler 480 (Roche) using iTaq Universal SYBR Green Supermix (Bio-Rad). Gene of interest were normalized to the 18S rRNA reference gene. Following gene-specific primer pairs were used: *S100A8*, forward: 5'-ATG-CCG-TCT-ACA-GGG-ATG-ACC-T-3', reverse: 5'-AGA-ATG-AGG-AAC-TCC-TGG-AAG-TTA-3'; *S100A9*, forward: 5'-GCT-TTG-ACA-GAG-TGC-AAG-ACG-ATG-AC-3', reverse: 5'-GAG-TAT-TGG-TGG-AAG-GTG-TTG-ATG-ATG-G-3'; *TLR4*, forward: 5'-CCC-TTC-TCA-ACC-AAG-AAC-CTG-GAC-3', reverse: 5'-CCC-ATC-TTC-AAT-TGT-CTG-GAT-TTC-ACA-C-3'; *TLR2*, forward: 5'-GCA-AGC-TGC-GGA-AGA-TAA-TG-3', reverse: 5'-CGC-AGC-TCT-CAG-ATT-TAC-CC-3'; *OCLDN*, forward: 5'-AGG-AGG-CCA-TTG-CCA-TTG-TAC-TG-3', reverse: 5'-TGC-TCA-TCA-TAA-ATG-TGT-TCC-TTG-TCC-C-3'; *18S*, forward: 5'-CGG-CTA-CCA-CAT-CCA-AGG-AA-3', reverse: 5'-AGC-TGG-AAT-TAC-CGC-GGC-T-3'.

GeneChip microarray assay

Sample preparation for microarray hybridization was performed as described in the Applied Biosystems GeneChip Whole Transcript Plus Reagent Kit user guide (Thermo Fisher Scientific, Waltham, MA). In brief, 200 ng of total RNA was used to generate double-stranded cDNA, from which 12 μ g of subsequently synthesized complementary RNA was purified

and reverse transcribed into single-stranded (ss) cDNA, in which unnatural deoxyuridine triphosphate residues were incorporated. Purified ss cDNA was fragmented using a combination of uracil DNA glycosylase and apurinic/apyrimidinic endonuclease 1 followed by a terminal labeling with biotin. Then, 3.8 µg of fragmented and labeled ss cDNA was hybridized to Applied Biosystems GeneChip Clariom S human arrays for 16 hours at 45°C and 60 rotations per minute in an Applied Biosystems GeneChip hybridization oven 640. Hybridized arrays were washed and stained in an Applied Biosystems GeneChip Fluidics Station FS450, and the fluorescent signals were measured with an Applied Biosystems GeneChip GeneChip Scanner 3000 7G system. Fluidics and scan functions were controlled by the Applied Biosystems GeneChip Command Console version 5.0 software. Sample processing was performed at a genomics Core Facility, Kompetenzzentrum Fluoreszenz Bioanalytik–Center of Excellence for Fluorescent Bioanalytics (Regensburg, Germany).

Microarray data were analyzed using Bioconductor packages³² in R version 4.3.1. During quality control, no batch effects were found; however, 1 outlier sample was identified and excluded from further analysis. Normalization was performed using the robust multiarray average algorithm from the oligo package,³³ and differential gene expression assessed by linear modeling using the limma package.³⁴

Neutrophil staining

Slides (2–3 µm thick) sectioned from formalin-fixed paraffin-embedded (FFPE) biopsy specimens were deparaffinized at 80°C for 30 minutes, immersed in xylene followed by descending alcohol line. Standard hematoxylin and eosin staining was performed with the addition of a naphthol AS-D chloroacetate esterase (CLAE) kit (Sigma-Aldrich, N0758), which specifically stains polynuclear granulocytes. Slides were then treated with an ascending alcohol line series followed by xylene, sealed with Entellan. The total number of neutrophil infiltrates were manually counted in 40× magnification in high-power field.

Data analyses

Data were analyzed in SPSS version 29 (IBM Corp) and R version 4.1.2. Correlations were depicted by Pearson or Spearman rho coefficients. *P* values were calculated with Mann-Whitney *U* test or Kruskal-Wallis test. Box plots were used to visualize continuous data. Factors associated with elevated calprotectin messenger RNA (mRNA) levels were assessed using logistic regression analysis. Receiver operating characteristics curves were generated and area under the curve was calculated to evaluate the outcome prediction. The Youden index was calculated to define calprotectin threshold level. Competing risk analysis was used to calculate outcomes whereas Cox regression was used for univariable and multivariable assessments.

Results

A microarray detects differential expression of neutrophil markers in aGI-GVHD

Total RNA from GI biopsy specimens was extracted and quantified using a microarray platform, resulting in measurements of 21 448 distinct mRNAs. When biopsy specimens were classified based on Lerner aGI-GVHD, granulocyte-associated genes

emerged as the most differentially expressed, as visualized in a volcano plot (Figure 1A; *n* = 97). Notably, the neutrophil-specific markers calprotectin (encoded by *S100A8/S100A9*) and CD16 (encoded by *FCGR3A/FCGR3B*) showed significant differential expression in aGI-GVHD, prompting a focused investigation of these markers. To account for the complexity of neutrophil-related gene expression,³⁵ we analyzed the relationship between mRNA levels of key neutrophil markers and the extent of neutrophil infiltration in matched biopsy samples. CLAE staining was conducted on a subset of FFPE paired biopsy specimens, including both non-GVHD (*n* = 14) and aGI-GVHD cases (*n* = 25). Among the 12 mRNAs that showed a strong correlation with neutrophil infiltration, calprotectin mRNA displayed the highest correlation, positioning it as a gene of particular interest (supplemental Table 2). Further analysis using a matrix scatterplot revealed that this correlation was primarily driven by the aGI-GVHD cohort rather than the non-GVHD group (Figure 1B), underscoring the relevance of neutrophil granulocytes in aGI-GVHD pathogenesis.

Neutrophil granulocyte calprotectin correlates with severity of aGI-GVHD

To investigate the role of neutrophils in aGI-GVHD, we quantified *S100A8* and *S100A9* mRNA expression using qPCR. *S100A8* and *S100A9* expression levels were examined in relation to the histological classification of aGI-GVHD, revealing a significant upregulation with increasing disease severity (Figure 2A; *n* = 575; *S100A8*, *P* < .001; *S100A9*, *P* < .001). This association suggests that *S100A8* and *S100A9*, markers for calprotectin, may reflect disease progression. Furthermore, we explored changes in calprotectin expression at clinical onset of aGI-GVHD. Analysis of onset biopsy specimens compared with disease-free screening biopsy specimens showed a marked increase in calprotectin expression at the onset of disease (Figure 2B; *n* = 407; *S100A8*, *P* < .001; *S100A9*, *P* = .006). Furthermore, onset biopsy specimens from steroid-refractory patients showed significantly higher calprotectin expression than steroid-sensitive patients (supplemental Figure 1; *n* = 180; *S100A8*, *P* = .001; *S100A9*, *P* = .004). In addition to gene expression, we analyzed mucosal (lamina propria) neutrophil infiltration in matched biopsy specimens using CLAE staining. Results demonstrated significantly higher degree of neutrophil infiltration with increasing aGI-GVHD severity (Figure 2C; *n* = 59; *P* < .001). Onset biopsy specimens also exhibited significantly elevated mucosal neutrophil infiltration compared with screening biopsy specimens (Figure 2D; *n* = 45, *P* = .005), underscoring that the presence of neutrophils closely correlates with disease onset and progression. An exemplary hematoxylin and eosin staining along with CLAE markers are shown for Lerner 0 and 4 biopsy specimens (Figure 2E). The Lerner 0 biopsy specimen showed intact epithelium and dense crypts without CLAE signals, whereas Lerner 4 biopsy specimen showed complete epithelial denudation with massive crypt abscess with strong CLAE signals in the lamina propria. Importantly, no neutrophil infiltration was detected in GVHD-free or screening biopsy specimens, indicating that neutrophil infiltration is associated with active aGI-GVHD.

TLR expression is associated with calprotectin

Based on the observed strong correlation between calprotectin levels and the severity of aGI-GVHD, we hypothesized

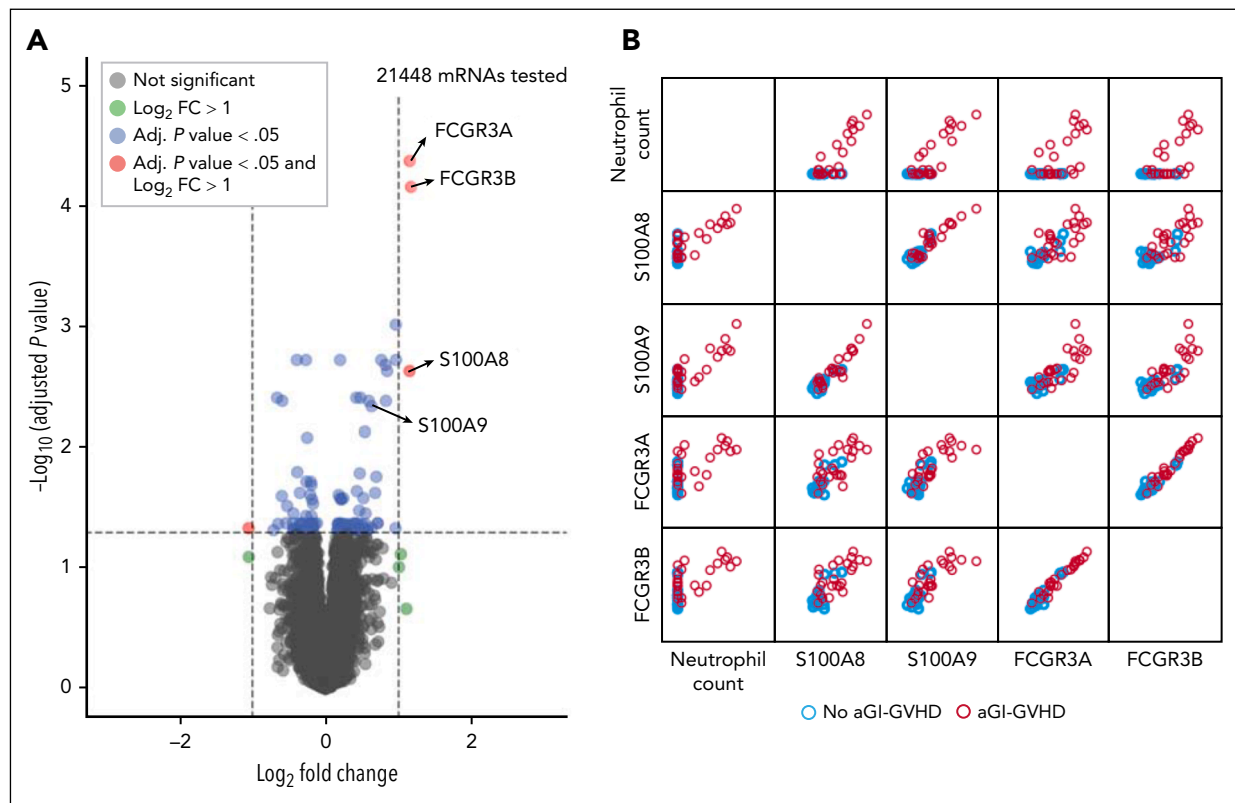


Figure 1. Differential expression of granulocyte markers in aGI-GVHD. (A) Volcano plot depicting gene expression differences between patients with and without aGI-GVHD, based on microarray analysis of 97 biopsy samples. Statistical adjustments were made for age, sex, and biopsy localization to control for potential confounding variables. Genes were classified as significantly differentially expressed if they exhibited a $\log_2FC \geq 1$ or ≤ -1 with an adjusted P value $< .05$. Notable genes of interest are indicated by black arrows. (B) Matrix scatter correlation plot illustrating the relationship between granulocyte marker expression and absolute neutrophil infiltration. It depicts pairwise associations among neutrophil infiltrates, *S100A8*, *S100A9*, *FCGR3A*, and *FCGR3B* mRNA expression levels in tissue biopsy specimens. Each scatterplot represents the relationship between 2 variables, with individual data points corresponding to expression levels in a given sample, analyzed in a subset of 39 matched biopsy specimens from panel A. Biopsy samples from patients with aGI-GVHD ($n = 25$) are denoted by red circles, whereas non-GVHD biopsy samples ($n = 14$) are marked with blue circles. Adj, adjusted; $\log_2FC$, $\log_2$ fold change.

that BT resulting from GVHD-mediated epithelial damage is a primary factor promoting neutrophil infiltration and subsequent calprotectin release. Bacterial recognition by the host is mediated via TLRs: TLR2 for gram-positive bacteria and TLR4 for gram-negative bacteria.³⁶ To investigate the potential link between TLR expression and calprotectin levels, we quantified mRNA expression of *TLR4* and *TLR2* and analyzed their association with calprotectin expression. Our findings revealed significantly elevated calprotectin levels in tissues with high *TLR4* expression compared to those with low *TLR4* expression. Specifically, *S100A8* expression was 24-fold higher and *S100A9* expression was 16-fold higher in *TLR4*-high biopsy specimens (Figure 3A; $n = 561$; $P < .001$). Similarly, high *TLR2* expression was associated with increased calprotectin levels, with *S100A8* showing a 14-fold increase and *S100A9* a ninefold increase relative to *TLR2*-low samples (Figure 3B; $n = 560$; $P < .001$). Additionally, matrix scatterplot revealed a strong and significant correlation between calprotectin subunits and TLRs (supplemental Figure 2).

Subsequent binary logistic regression analysis was conducted to assess the associated factors for elevated calprotectin expression. Independent variables considered classical risk factors for ASCT outcomes were first analyzed in a univariable manner. Only those that showed significant risk were included in a

multivariable model. Results confirmed that severe aGI-GVHD (*S100A8* [odds ratio, 2.8; 95% confidence interval (CI), 1.7-4.8; $P < .001$] and *S100A9* [odds ratio, 2.0; 95% CI, 1.2-3.3; $P = .004$]), steroids (*S100A8* [odds ratio, 2.1; 95% CI, 1.4-3.3; $P < .001$] and *S100A9* [odds ratio, 2.6; 95% CI, 1.7-4.0; $P < .001$]), and high *TLR4* expression (*S100A8* [odds ratio, 7.9; 95% CI, 4.9-12.8; $P < .001$]; *S100A9* [odds ratio, 6.9; 95% CI, 4.3-11.2; $P < .001$]) were independent associated factors for elevated calprotectin levels (Table 1). Although *TLR2* did not reach statistical significance, it exhibited a trend toward being independently associated with *S100A8* expression (*S100A8* [odds ratio, 1.4; 95% CI, 0.9-2.3; $P = .09$] and *S100A9* [odds ratio, 1.1; 95% CI, 0.7-1.8; $P = .4$]). These findings strongly suggest that TLR expression plays a critical role in driving calprotectin expression.

Calprotectin expression differs based on the anatomical location of the GI tract

Murine GVHD studies link neutrophil influx mainly to the small intestine,^{28,29} whereas human studies associate fecal calprotectin with the colon.¹⁴ To clarify this discrepancy, we analyzed calprotectin expression in the UGI and LGI. Calprotectin levels were significantly higher in the LGI, with *S100A8* and *S100A9* expression elevated 2.7-fold ($P = .008$) and 2.4-fold ($P = .08$), respectively (Figure 4A). This finding corresponded with increased *TLR4* expression ($P = .03$) in the LGI (Figure 4B).

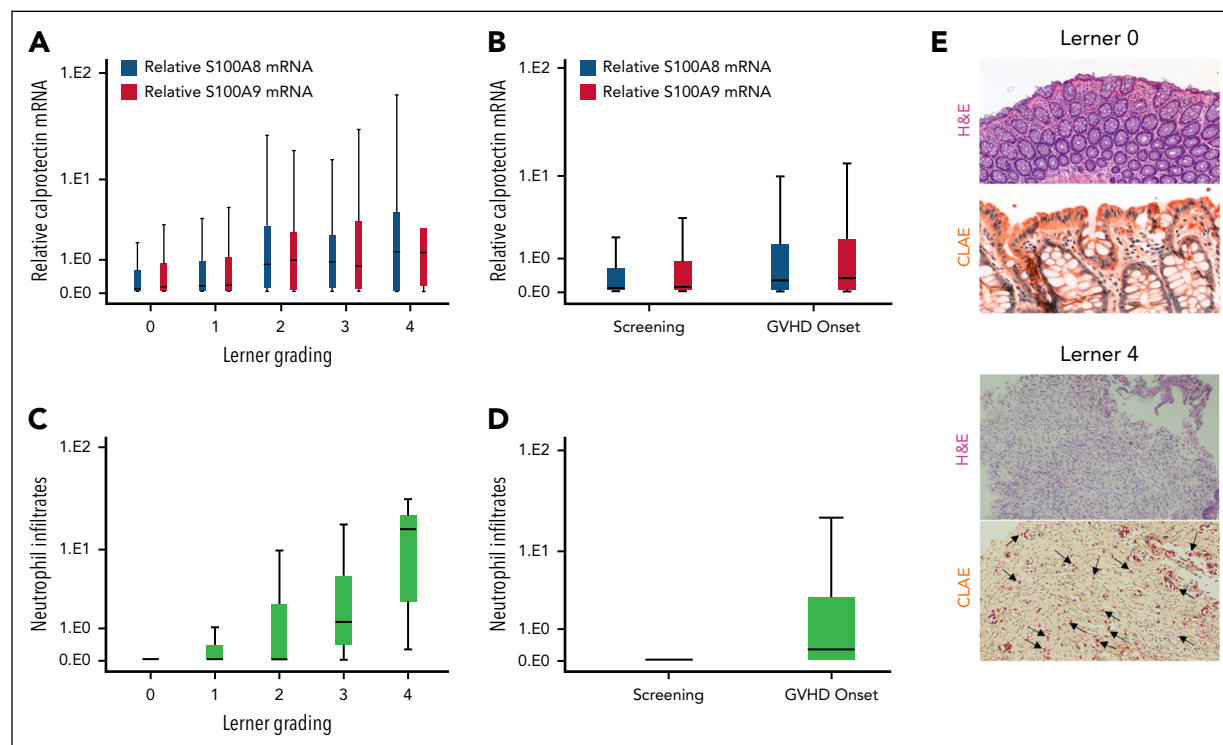


Figure 2. Calprotectin expression upregulates at aGI-GVHD onset and correlates with disease severity. (A) Calprotectin mRNA expression across Lerner classification stages of aGI-GVHD (N = 575 [4 biopsy specimens could not be graded]; $P < .001$ for both S100A8 and S100A9 subunits). The x-axis represents histologically defined aGI-GVHD severity, and the y-axis represents mRNA expression in a log scale. (B) Calprotectin mRNA expression in a clinical classification comparing disease-free “screening” biopsy specimens vs at GVHD onset (n = 407; $P < .001$ for S100A8 and $P = .006$ for S100A9). The x-axis represents clinically defined GVHD, and y-axis represents mRNA expression in a log scale. (C) Neutrophil infiltration across Lerner classification stages of aGI-GVHD (n = 59; $P < .001$). Note: Lerner 0 biopsy specimens exhibited no detectable neutrophils. The x-axis represents histologically defined aGI-GVHD, and the y-axis represents number of neutrophil infiltrates per high-power field in log scale. (D) Neutrophil infiltration in biopsy specimens clinically classified as disease-free “screening” biopsy specimens vs at aGI-GVHD onset (n = 45; $P = .005$). The x-axis represents clinical classification of GVHD, and the y-axis represents extent of neutrophil infiltration in high-power field in a log scale. (E) Representative histological images: H&E and CLAE staining of Lerner 0 and Lerner 4 biopsy specimens. The Lerner 0 biopsy specimen displays intact epithelium, abundant healthy crypts, and an absence of neutrophils. In contrast, the Lerner 4 biopsy specimen shows complete epithelial denudation, crypt abscess formation and extensive neutrophil infiltration. Black arrows show infiltrated granulocytes in the lamina propria. H&E, hematoxylin and eosin.

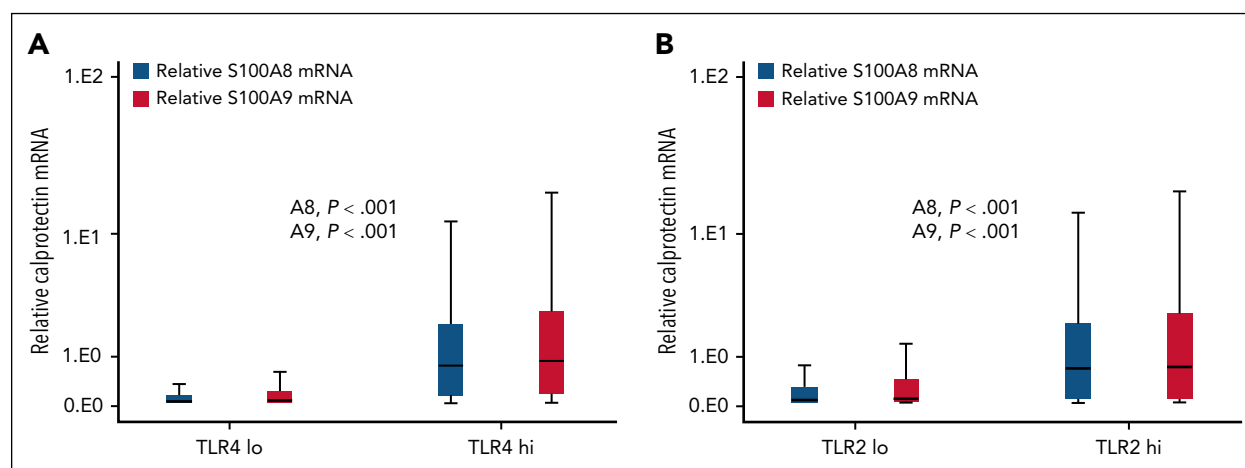


Figure 3. TLR signaling is associated with increased calprotectin expression in GI tissue. (A) mRNA expression levels of calprotectin subunits S100A8 and S100A9 in biopsy specimens stratified by TLR4 expression. Tissues with high TLR4 expression exhibit significantly elevated S100A8 and S100A9 levels compared to TLR4-low tissues (N = 561; $P < .001$ for both). (B) mRNA expression of S100A8 and S100A9 in biopsy specimens categorized by TLR2 expression. Similarly, high TLR2 expression is associated with increased calprotectin levels (N = 560; $P < .001$ for both). In each graph, the x-axis represents stratified TLR expression, whereas the y-axis represents calprotectin mRNA expression in log scale. hi, high; lo, low.

Table 1. Logistic regression analysis for parameters correlating with calprotectin subunits after ASCT (significant parameters of univariable testing)

All biopsy specimens	S100A8		S100A9	
	P value	Odds ratio (95% CI)	P value	Odds ratio (95% CI)
aGI-GVHD (low vs high)	<.001	2.86 (1.70-4.81)	.004	2.05 (1.25-3.36)
Steroids (≤ 20 mg vs > 20 mg)	<.001	2.19 (1.44-3.35)	<.001	2.66 (1.76-4.02)
TLR4 (low vs high)	<.001	7.95 (4.91-12.88)	<.001	6.95 (4.30-11.23)
TLR2 (low vs high)	.09	1.49 (0.93-2.39)	.49	1.17 (0.73-1.88)

Calprotectin was further analyzed in low and high Lerner score groups. In severe GVHD, *S100A8* and *S100A9* increased 30-fold and 23-fold in the UGI ($n = 214$; $P < .001$), but only five-fold and threefold in the LGI ($n = 361$; *S100A8*, $P < .001$; *S100A9*, $P = .02$), suggesting a stronger net change in the UGI. Notably, in low Lerner cases, calprotectin was higher in the LGI ($n = 451$; *S100A8*, $P = .001$; *S100A9*, $P = .02$), reflecting microbial exposure. However, in severe GVHD, calprotectin was higher in the UGI than in the LGI ($n = 124$; *S100A8*, $P = .04$; *S100A9*, $P = .1$), indicating that the LGI may have greater microbial tolerance and a dampened inflammatory response

(Figure 4C). To account for severe epithelial and mucosal depletion in severe aGI-GVHD, we normalized mucosal calprotectin expression to *OCLDN* gene that codes for tight junction protein occludin, which is critical for maintaining epithelial barrier integrity. The observed distinction between UGI and LGI with respect to GVHD severity persisted in the normalized data (supplemental Figure 3).

Calprotectin expression associates with TRM

Following the observed differences between small and large bowel, we aimed to investigate the prognostic capability of

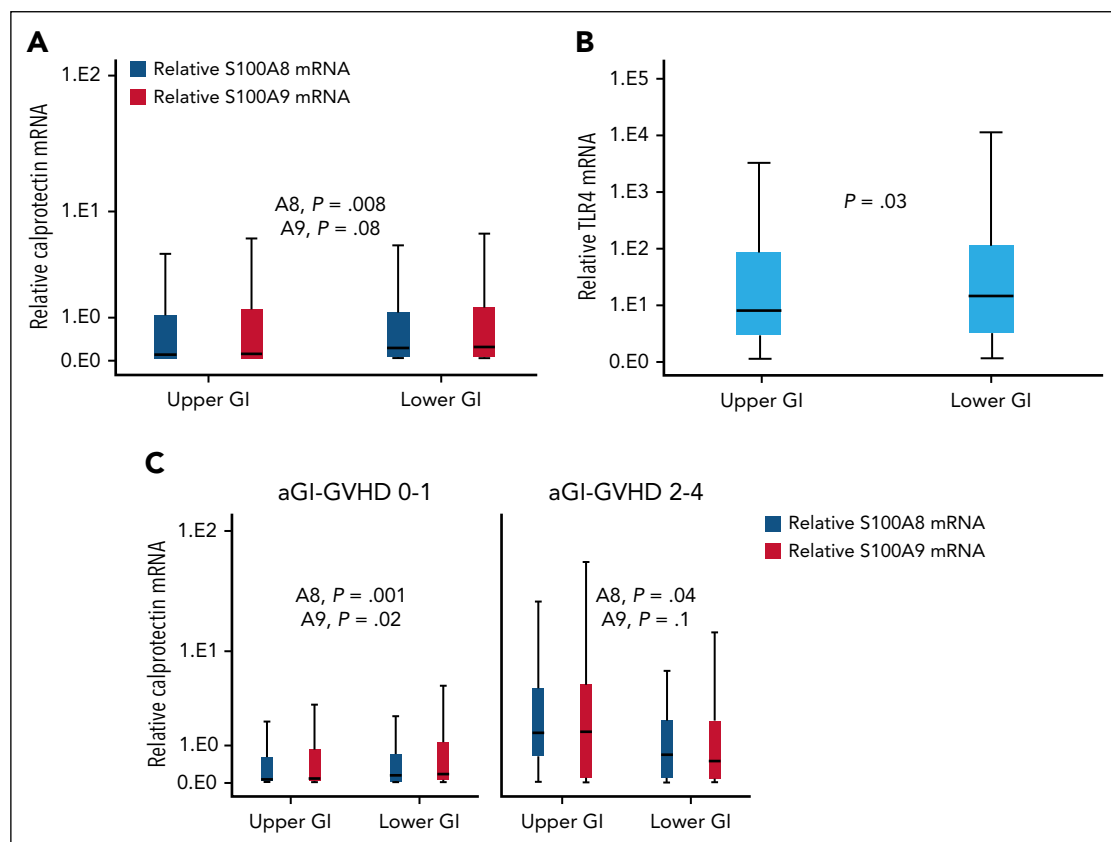


Figure 4. Calprotectin expression varies between the small and large intestines. (A) Calprotectin mRNA expression (*S100A8* and *S100A9*) differs between the UGI and LGI, with statistical significance observed for *S100A8* ($N = 579$; $P = .008$) and a strong trend for *S100A9* ($P = .08$). (B) *TLR4* mRNA expression also differs between UGI and LGI ($N = 566$, $P = .03$). (C) In low Lerner score biopsy specimens, calprotectin mRNA expression is significantly different between UGI and LGI ($n = 451$; $P = .001$ for *S100A8* and $P = .02$ for *S100A9*). During inflammation (high Lerner score: aGI-GVHD 2-4), a significant difference is observed for *S100A8* ($n = 124$; $P = .04$), whereas *S100A9* shows a trend that does not reach statistical significance ($P = .1$). In each graph, the x-axis represents anatomical location of biopsy, and the y-axis represents mRNA expression in log scale.

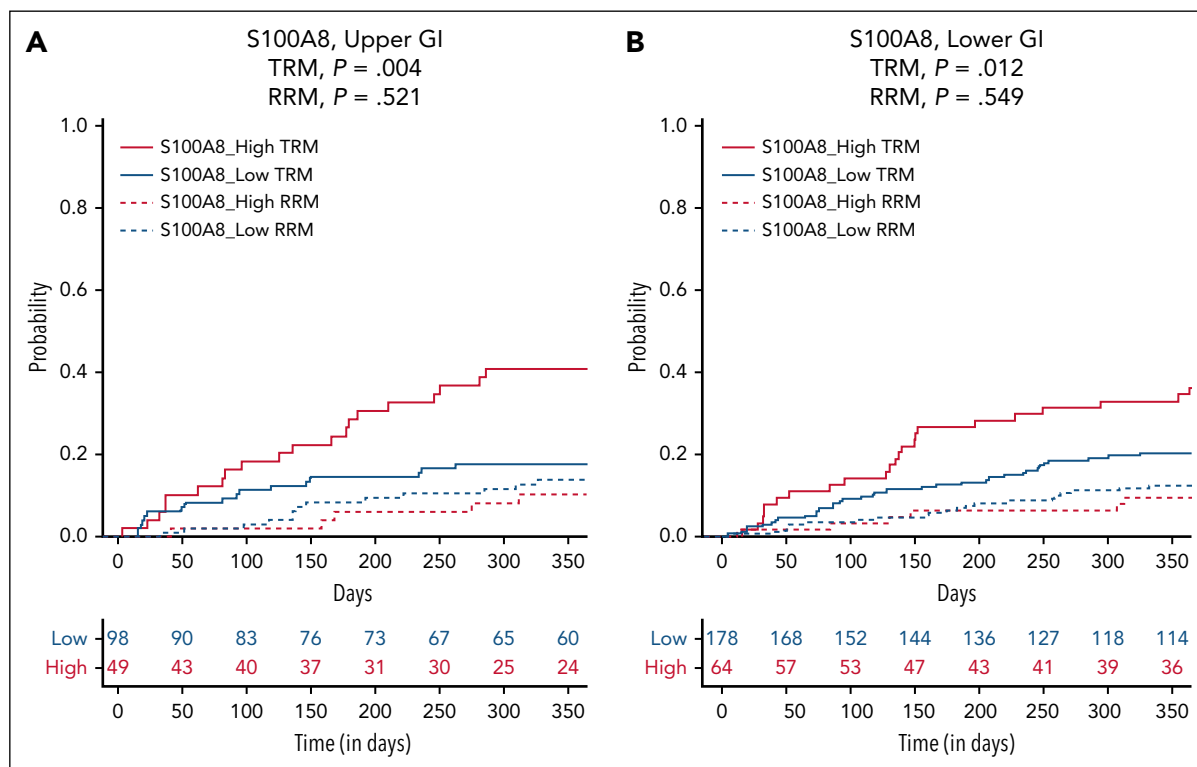


Figure 5. Calprotectin expression associates with TRM without affecting RRM. (A) Competing risk analysis curve for the *S100A8* subunit in the UGI ($n = 147$; $P = .004$ for TRM and $P = .521$ for RRM). (B) Competing risk analysis curve for the *S100A8* subunit in the LGI ($n = 242$; $P = .012$ for TRM and $P = .549$ for RRM). The x-axis represents time in days after biopsy retrieval, and the y-axis represents probability of death. For each curve, the number of patients at risk at each time point is provided below the respective graph.

tissue calprotectin in small and large bowel separately. In the context of ASCT, relapse-related mortality (RRM) is a competing event for TRM; therefore, competing risk analyses was performed. Interestingly, in UGI and in LGI, calprotectin subunit *S100A8*, but not *S100A9*, clearly showed a significant association with 1-year TRM, whereas there was no difference in RRM (Figure 5A [TRM, $P = .004$; RRM, $P = .52$]; Figure 5B [TRM, $P = .01$; RRM, $P = .54$]). Finally, we performed Cox regression analysis to evaluate the factors that affect 1-year TRM. In the UGI, in univariable analysis, aGI-GVHD (hazard ratio [HR], 4.12; 95% CI, 2.13-7.97; $P < .001$) and high level of calprotectin (*S100A8* [HR, 2.47; 95% CI, 1.29-4.72; $P = .006$] and *S100A9* [HR, 1.98; 95% CI, 1.03-3.77; $P = .03$]) were an independent risk factors. Similarly, in the LGI, aGI-GVHD (HR, 2.81; 95% CI, 1.67-4.72; $P < .001$) and high calprotectin expression (*S100A8* [HR, 1.95; 95% CI, 1.15-3.30; $P = .013$] and *S100A9* [HR, 2.27; 95% CI, 1.18-4.39; $P = .014$]) were observed to be an independent risk factor. In multivariable analysis, aGI-GVHD alone remained the risk factor for TRM in both in the UGI and the LGI (UGI [HR, 3.28; 95% CI, 1.50-7.19; $P = .003$] and LGI [HR, 2.55; 95% CI, 1.50-4.33; $P < .001$]; Table 2).

Broad-spectrum antibiotics suppress calprotectin expression

Our findings demonstrated that calprotectin expression was directly associated with the onset and severity of aGI-GVHD, with BT likely serving as a major trigger for neutrophil infiltration and, consequently, elevated calprotectin levels. Based on this, we hypothesized that managing BT could modulate calprotectin

expression. To test this, we first analyzed the kinetics of antibiotic exposure duration on mucosal calprotectin expression (supplemental Figure 4) and then classified biopsy specimens into “antibiotics” and “no antibiotics” groups. We then analyzed aGI-GVHD onset patients who received broad-spectrum antibiotics (piperacillin/tazobactam or meropenem/vancomycin) were compared to those who did not receive antibiotics. Remarkably, biopsy samples from patients treated with broad-spectrum antibiotics exhibited significantly lower calprotectin expression than those not receiving antibiotics (Figure 6; *S100A8*, $n = 181$ [$P = .001$]; *S100A9*, $n = 181$ [$P = .012$]). This observation supports the conclusion that elevated calprotectin levels in aGI-GVHD, which are likely due to BT, may be effectively mitigated by antibiotic treatment. Strikingly, the antibiotics group showed a pronounced shift toward steroid sensitivity (67.5% sensitive vs 32.5% refractory; supplemental Figure 5).

Discussion

To our knowledge, this study is the first to leverage a substantial collection of unique GI mucosal biopsy specimens to elucidate the correlation between neutrophil granulocytes and the severity of aGI-GVHD, as well as to demonstrate the prognostic value of GI calprotectin. Our findings indicate a role of BT in activation of neutrophils explaining beneficial effects of broad-spectrum antibiotics when given at the onset of aGI-GVHD. In addition, we identified significant differences in pathophysiological mechanisms between the small and large intestines, highlighting the site-specific nature of aGI-GVHD.

Table 2. Cox regression analysis for 1-year TRM

Variables	P value	HR (95% CI)	P value	HR (95% CI)
	UGI		LGI	
Univariable				
Lerner (low vs high)	<.001	4.12 (2.13-7.97)	<.001	2.81 (1.67-4.72)
S100A8	.006	2.47 (1.29-4.72)	.013	1.95 (1.15-3.3)
S100A9	.030	1.98 (1.03-3.77)	.014	2.27 (1.18-4.39)
Multivariable				
Lerner (low vs high)	.003	3.20 (1.50-7.19)	<.001	2.55 (1.50-4.33)
S100A8	.45	1.43 (0.55-3.68)	.33	1.38 (0.71-2.69)
S100A9	.87	1.07 (0.44-2.55)	.33	1.49 (0.66-3.39)

Both univariable and multivariable testing are shown for small and large intestines. $P < .05$ is considered significant (bold).

In this study, we identified granulocyte markers were among the most differentially regulated genes in aGI-GVHD. Correlating granulocyte gene expression with total neutrophil influx in matched FFPE biopsy specimens revealed that the calprotectin subunits S100A8 and S100A9 exhibit the strongest correlation. Calprotectin has been widely recognized for over 3 decades as a marker predominantly associated with activated neutrophils,^{22,37} making it a robust indicator for studying neutrophil-mediated inflammation. Notably, fecal calprotectin has been extensively studied and validated as a predictive biomarker in inflammatory bowel disease^{10,11,38,39} and GVHD,^{14,26,27} with previous studies demonstrating a strong correlation between stool calprotectin levels and intestinal inflammation. Consistent with previous findings, we observed significant upregulation of calprotectin subunits with increasing severity of aGI-GVHD. Moreover, calprotectin levels rose markedly at the onset of aGI-GVHD, especially in steroid-refractory cases, which aligns

with recent findings by Patel et al.⁴⁰ Complementing these results, neutrophil infiltration into tissues also correlated strongly with GVHD severity. Importantly, in the absence of aGI-GVHD, neutrophil infiltration was absent. This observation led us to hypothesize that tissue damage is a prerequisite for neutrophil infiltration. We propose that aGI-GVHD recruit neutrophils to sites of tissue damage, at which point they contribute to exacerbating the disease through further tissue injury. BT across the gut mucosa, a consequence of GVHD-induced barrier disruption, is likely a key driver of neutrophil infiltration. This hypothesis aligns with findings by Schwab et al, who demonstrated in a murine model of GVHD that neutrophil recruitment to the intestine was primarily driven by BT from the gut lumen to peri-intestinal tissues.²⁸

Bacterial pathogen-associated molecular patterns are recognized by host cells via pathogen recognition receptors, such as TLRs. Specifically, gram-positive and gram-negative bacteria are predominantly recognized by TLR2 and TLR4, respectively.^{36,41,42} This mechanism facilitates the host's immune response to BT. Given the known recruitment of neutrophils to target organs during BT, we hypothesized that the TLR pathway plays a central role in calprotectin-mediated inflammation in aGI-GVHD. Supporting this hypothesis, calprotectin subunit levels were found to be 16 to 24 times higher in patients with elevated *TLR4* expression and 9 to 14 times higher in those with elevated *TLR2* expression after ASCT. Consistent with our findings, prior studies have shown that neutrophils lacking TLRs result in less severe GVHD.²⁸ Furthermore, a regression analysis identified aGI-GVHD, steroids and *TLR4* as an independent associated factor for elevated calprotectin levels, whereas association with *TLR2* was less pronounced. The preference for TLR4 signaling may be attributed to its role as an endogenous receptor for calprotectin and the predominance of gram-negative BT in aGI-GVHD.^{24,43} This underscores the importance of TLR4 in linking BT to neutrophil-mediated tissue damage in GVHD. Our findings, however, suggest but do not prove a TLR-calprotectin signaling axis. Further analyses assessing tissue microbiota, tissue bacterial load, or tissue lipopolysaccharide, as well as in vitro correlative studies and in vivo TLR modulation, are needed to fully confirm the hypothesis.

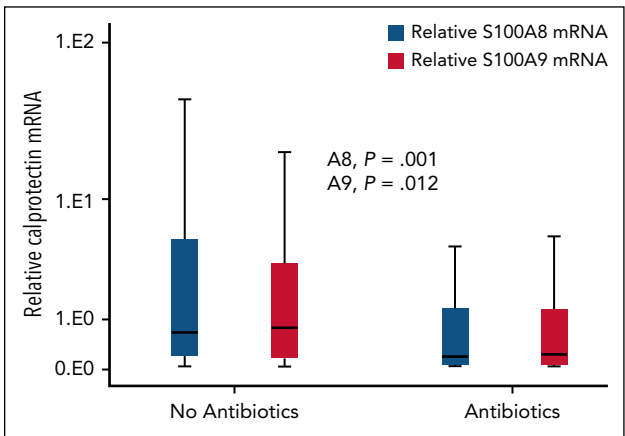


Figure 6. Impact of broad-spectrum antibiotics on calprotectin expression at the onset of aGI-GVHD. Patients received broad-spectrum antibiotics due to fever and infection. To assess the effect of antibiotic exposure on calprotectin expression, patient biopsy specimens at disease onset were categorized into 2 groups. The "no antibiotics" group ($n = 47$) included biopsy specimens from patients who were not exposed to antibiotics or had an exposure duration of <6 days. The "antibiotics" group ($n = 134$) consisted of biopsy specimens from patients who had received antibiotics for >6 days and/or whose antibiotic treatment ended within the preceding 14 days ($n = 181$; $P = .001$ for S100A8 and $P = .012$ for S100A9).

Of interest, steroids showed up to be a risk factor for calprotectin. Although steroids are given as a first-line therapy for GVHD, they are known to cause neutrophilia by (1) stimulating the bone marrow to release more neutrophils, (2) promoting demargination from endothelial lining,⁴⁴ and (3) delaying the apoptosis.⁴⁵ This effect of steroid may favor relatively long-lived neutrophil, contributing to higher calprotectin levels. Nevertheless, detailed and careful studies are needed to fully address the role of steroids.

One of the most striking findings of our study was the differential expression of calprotectin across distinct anatomical regions of the GI tract. In the homeostatic state or in the low Lerner group, the LGI exhibited higher calprotectin levels compared to the UGI, reflecting the higher microbial load in the LGI that necessitates a robust antimicrobial defense mechanism to suppress infection, leading to elevated calprotectin levels, and the source of calprotectin are possibly macrophages,⁴⁶ monocytes,⁴⁷ and myeloid derived suppressor cells.⁴⁸ However, during aGI-GVHD (particularly the high Lerner group), the UGI displayed markedly higher increase in calprotectin expression than the LGI. Specifically, the net increase in calprotectin levels during aGI-GVHD was 23 to 30-fold in the UGI and threefold to fivefold in the LGI. We assume that the LGI likely induces regulatory pathways,^{49,50} possibly mediated by the higher abundance of commensal microbiota, to counteract ongoing inflammation which limits the net increase in calprotectin levels. Conversely, the UGI exhibits heightened sensitivity to microbial antigens, potentially because of the high density of immune cells residing in this region, leading to enhanced immune surveillance and a greater increase in calprotectin expression during aGI-GVHD.^{51,52} These findings align with previous studies demonstrating significant neutrophil infiltration in the small intestine but not in the large intestine during aGI-GVHD.^{28,29}

Clinically, we observed an association of elevated calprotectin levels with worse outcomes. Competing risk models revealed that higher mucosal calprotectin levels were a significant predictor of survival disadvantage. In the UGI and the LGI, calprotectin emerged as an independent risk factor for TRM in a univariate Cox proportional hazards model. However, in a multivariable model, aGI-GVHD alone remained the dominant risk factor for TRM, reflecting high correlation between calprotectin levels and GVHD severity. These findings suggest that although GI-GVHD is the primary trigger for neutrophil influx and elevated calprotectin expression, the detrimental effect of calprotectin on TRM is likely secondary to the inflammatory cascade initiated by GI-GVHD. GVHD-mediated translocation of bacterial species may further aggravate neutrophil and T-cell activation, causing irreparable damage to the intestine.

Given the central role of BT in GVHD pathology,²⁸ we explored whether broad-spectrum antibiotics could affect calprotectin levels. Remarkably, patients who received broad-spectrum antibiotics at disease onset exhibited significantly reduced calprotectin levels. We propose that antibiotics lower BT by reducing bacterial load, thereby decreasing neutrophil infiltration and associated inflammation. Despite the well-documented adverse effects of antibiotics on ASCT outcomes,⁵³⁻⁵⁵ and strong recommendations to limit their use in ASCT recipients,⁵⁶ our findings highlight a paradox: beneficial effect of antibiotic

treatment in the context of aGI-GVHD might depend on timing. In murine GVHD models, germ-free or antibiotic-treated mice demonstrated reduced neutrophil influx and improved survival compared to untreated controls.^{28,29} It is, therefore, logical to hypothesize that patients diagnosed with aGI-GVHD might benefit from antibiotics by controlling neutrophil recruitment, and consequently decreasing inflammation.

With an extensive data set of intestinal biopsy specimens, our findings strongly highlight the diagnostic and prognostic value of mucosal calprotectin in aGI-GVHD. Whether calprotectin expression can enhance the prognostic sensitivity of biopsy samples requires evaluation. Although the primary goal of this study was to address a key gap in the field, direct assessment of calprotectin expression in intestinal biopsies, which had not been systematically investigated, fecal/serum calprotectin should also be prospectively investigated as a noninvasive biomarker. We identify substantial immunological differences between the UGI and LGI, providing insights into their respective immune responses. Moreover, these findings highlight a strategic therapeutic window for antibiotic interventions, where the benefits could surpass the potential risks. Although gut microbiome dysbiosis has been associated with BT and bacteremia,⁵⁷ it is still unclear what is cause and effect. Therefore, carefully designed clinical trials, potentially combining short-term antibiotics with subsequent restoration of microbiota by fecal microbiota transplant, may help answer these questions.

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Authorship

Contribution: E.E.G.K., E.M., and J.F. performed research, collected and analyzed data, and reviewed the manuscript; K.C. and A.H. performed statistical analysis; A.M. and D.H. performed research; M.H., A.-S.K., C.M., S.B., E.T.O., P.B., A.K., M. Evert, K.H., M.K., M. Edinger, D. Wolff, W.H., H.P., and A.G. analyzed and interpreted data, provided critical comments, and reviewed the manuscript; D. Weber collected data, analyzed and interpreted data, and reviewed the manuscript; B.K. performed statistical analysis, analyzed and interpreted data, and reviewed the manuscript; E.H. designed research, collected data, analyzed and interpreted data, performed statistical analysis, and reviewed the manuscript; and S.G. designed and performed research, collected data, analyzed and interpreted data, performed statistical analysis, and wrote the manuscript.

Conflict-of-interest disclosure: E.H. is a member of the scientific advisory boards of Matt Pharma (Lyon, France) and PharmaBiome (Zürich, Switzerland). The remaining authors declare no competing financial interests.

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Footnotes

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